

Non-Scalar Gradient Content in Few-Step Consistency Training: A Paired Counterfactual Measurement*

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Abstract

Few-step generative training commonly assumes that a training intervention—a change in the discretization gap—is *scalar-equivalent* to a learning-rate change, i.e. absorbable by a single scalar multiplier on the parameter update. We test this assumption in the optimizer-update space, conditioned on a fixed, nontrivial RAdam state, using a paired counterfactual audit (identical minibatch, timestep, noise, dropout; gap $g = 1.0$ vs. $g = 1.3$).

We find that the scalar moment-memory chain—an exact identity relating the coordinate-wise update-ratio gauge to the gradient-scale history—predicts $h \approx 1.001$, but the actual update ratio is $h \approx 0.837$, driven by a non-scalar per-step gradient residual. The residual is reproducible and structured: it grows with layer depth and is a monotone function of $|g - 1|$, but is not magnitude-driven. Crucially, the residual is *not* harmful: the arm carrying it ($g = 1.3$) achieves dramatically better FID/KID than the zero-residual arm ($g = 1.0$) at both NFE=1 and NFE=2 (FID $-34\%/-36\%$, KID $-37\%/-45\%$).

We conclude that scalar-equivalence reasoning in few-step training is systematically violated by non-scalar gradient content—a phenomenon the field’s scale-invariance literature (which analyzes only scalar/global rescaling) does not cover—but that this content is benign, not a source of quality degradation. We provide a reproducible paired-counterfactual measurement and a structural characterization of where and why scalar equivalence fails.

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*Draft framework for ICLR 2027. Role C (theory lead). All numbers from the repository’s own audits/evaluations; see ‘analysis/’.

1 Introduction

The scalar-equivalence assumption: a schedule/gap change is often treated as a learning-rate change. We ask whether it holds in the optimizer-update space, conditioned on a fixed optimizer state. The honest finding: it fails, but benignly.

2 Setup and method

Rectified RAdam, coordinate-wise update-ratio gauge

$$h_{k,i} = \frac{U_{g,k,i}}{U_{1,k,i}}, \quad i \in \mathcal{S}_k := \{i : U_{1,k,i} \neq 0\}. \quad (1)$$

The moment-memory identity as a falsifiable null model:

$$h_{k,i} = \frac{(1 + A_{k,i}^{(1)})}{\sqrt{1 + 2A_{k,i}^{(2)} + B_{k,i}^{(2)}}}. \quad (2)$$

The paired counterfactual audit (identical randomness; $g = 1.0$ vs. $g = 1.3$). Measured audit numbers: $a_K^* = 0.761$, $R_{\text{grad}} = 5.85\%$, $R_{\text{opt}} = 8.57\%$, $H_K = R_{\text{opt}}$ exact (energy gap 1.7×10^{-18}), $R_{\text{opt}} - R_{\text{grad}} = +0.027$.

3 Results: the diagnosis

3.1 The scalar null is falsified

Predicted $h \approx 1.001$ (median) vs. actual $h \approx 0.837$ (std 0.058), $\text{Corr} \approx 0$ (both stably different, not noisy).

3.2 The residual is structured, not noise (E5)

Gauge deviation from 1 grows with layer depth (pearson +0.46); not magnitude-correlated; gradient direction residual is a monotone function of $|g - 1|$.

3.3 The residual is not harmful (E6 + clean FID)

$g = 1.3$ (max residual) vs. $g = 1.0$ (zero residual): FID-5k $-34\%/-36\%$, KID-5k $-37\%/-45\%$ at NFE=1/2. $\text{corr}(\text{residual}, \text{FID}) \approx -0.99$: larger residual $\rightarrow$ better FID. The zero-residual arm ($g = 1.0$) is the worst.

3.4 The residual is state-conditioned (E7)

The gradient residual R_{grad} grows with optimizer maturity (n_K): 0.031 ($n_K = 243$) $\rightarrow$ 0.091 ($n_K = 1991$), roughly monotone. The update distortion R_{opt} is small (0.012–0.017) and $R_{\text{opt}} - R_{\text{grad}}$ is negative (the optimizer compresses the gradient residual). Measurement-sensitivity caveat: R_{opt} magnitude and the sign of $R_{\text{opt}} - R_{\text{grad}}$ depend on the `loss_fn` checkpoint / minibatch (earlier audit: $R_{\text{opt}} = 0.086$, $+0.027$; here: 0.014 , -0.076 at the same n_K).

3.5 Robustness (E11)

Cross-seed: at 1024k, the larger-gap arm ($g = 1.1$) beats $g = 1.0$ on FID-50k across all 3 seeds (5/6 cells). Support-threshold: $\text{Corr} \approx 0$ / scalar-null-falsified is stable across $\text{atol } 10^{-5}/10^{-6}$.

4 Related work and positioning

- $\beta_1=\beta_2$ gradient-scale invariance (global scalar; we are coordinate-wise, state-conditioned, non-scalar).
- Adam-atan2 / DeVA (scalar rescaling decomposition; we analyze non-scalar content).
- ADCM (schedule for quality; we analyze optimizer-state equivalence).
- SCT (objective/variance theory; we analyze update geometry).
- Dead-Direction Conditioners (symmetry-orbit gauge; we rename ours to “coordinate-wise update-ratio gauge”).

5 Discussion and limitations

$g = 1.3$ differs in the whole gap (scalar + non-scalar), so we do not claim the residual *causes* the FID improvement—only that it is not harmful. FID-5k (not FID-50k), 256-kimg runs, seed 3, single schedule pair. The residual-magnitude measurement convention varies (3.2% from 20-step replay vs. 5.85%/8.57% from single-step audit); we report the audit values. The 3.2% $\rightarrow$ 16% attribution ($1.001 \rightarrow 0.837$) is not derived; we report the measured h_{actual} directly.

6 What this paper does not claim

- NOT “the residual is harmful” (falsified: $g = 1.3$ wins FID).
- NOT “a residual-corrected update improves quality” (contraindicated).
- NOT “first adaptive discretization / optimizer invariance / history-gauge theorem” (pre-empted).
- NOT “gap is equivalent to learning-rate change” (falsified by $h = 0.837$).

7 Measured numbers

quantity	value	source
a_K^* (gradient scalar fit)	0.761	stateful audit
R_{grad}	5.85%	stateful audit
R_{opt}	8.57%	stateful audit
$H_K = R_{\text{opt}}$	exact (1.7×10^{-18})	stateful audit
$h_{\text{predicted}}$ (scalar null)	1.001	moment-memory
h_{actual}	0.837 (std 0.058)	moment-memory result
$\text{Corr}(h_{\text{pred}}, h_{\text{actual}})$	≈ 0	moment-memory result
gauge dev vs. depth (pearson)	+0.46	E5
corr(residual, FID)	−0.99	E6
$g = 1.3$ vs. $g = 1.0$ FID-5k	−34%/−36% (NFE1/2)	clean FID
$g = 1.3$ vs. $g = 1.0$ KID-5k	−37%/−45% (NFE1/2)	clean FID
R_{grad} vs. n_K (243 $\rightarrow$ 1991)	0.031 $\rightarrow$ 0.091	E7
cross-seed (1024k, $g = 1.1$)	wins 5/6 FID-50k cells	E11
$\text{Corr} \approx 0$ across atol $10^{-5}/10^{-6}$	0.005 / 0.0002	E11